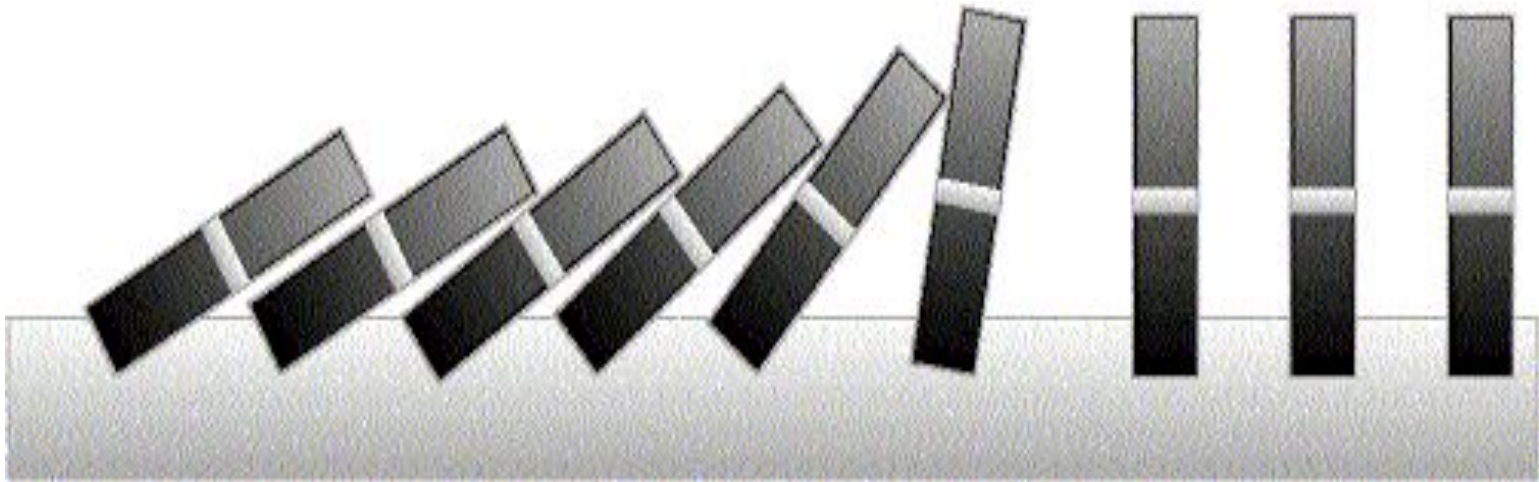


# Mathematical Induction I



# This Lecture

We have already discussed different proof techniques.

This time we will focus on probably the most important one

- mathematical induction.

This lecture's plan is to go through the following:

- The idea of mathematical induction
- Basic induction proofs (e.g. equality, inequality, property, etc)
- An interesting example
- A paradox

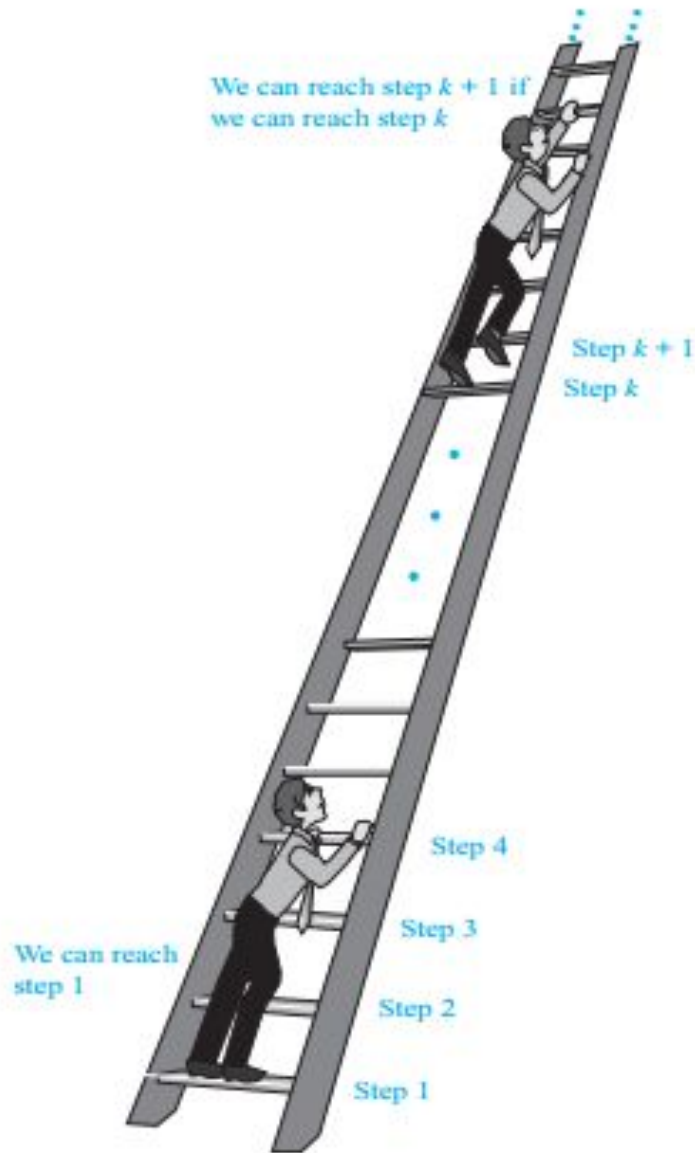
# Why Induction

- We need to prove many statements that are true for a set (e.g., all integers):
  - For every positive integer  $n$ :
    - $n! \geq n^n$
    - $n^3 - n$  is divisible by 3
    - A set with  $n$  elements has  $2^n$  subsets
    - The sum of first  $n$  positive integers is  $n(n+1)/2$
    - etc.

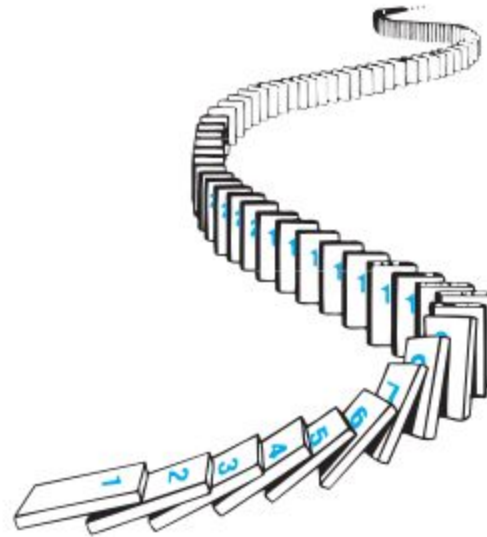
# Induction: Main Idea

- Proofs using mathematical induction have two parts (assuming the domain of positive integer).
  - First, they show that the statement holds for the positive integer 1.
  - Second, they show that if the statement holds for a positive integer then it must also hold for the next larger integer.
- Formally this is based on the rule of inference:
  - if  $P(1)$  and  $\forall k(P(k) \rightarrow P(k + 1))$  are true for the domain of positive integers, then  $\forall n P(n)$  is true.

# Main Idea of Induction



**FIGURE 1** Climbing an Infinite Ladder.



**FIGURE 2** Illustrating How Mathematical Induction Works Using Dominoes.

# Principle of MI

*PRINCIPLE OF MATHEMATICAL INDUCTION* To prove that  $P(n)$  is true for all positive integers  $n$ , where  $P(n)$  is a propositional function, we complete two steps:

*BASIS STEP:* We verify that  $P(1)$  is true.

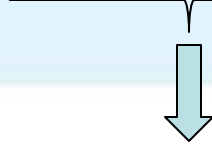
*INDUCTIVE STEP:* We show that the conditional statement  $P(k) \rightarrow P(k + 1)$  is true for all positive integers  $k$ .

# Principle of MI

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*INDUCTIVE STEP:* We show that the conditional statement  $P(k) \rightarrow P(k+1)$  is true for all positive integers  $k$ .



If  $P(k)$  is true then  $P(k+1)$  is true

# Principle of MI

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Induction Hypothesis

If  $P(k)$  is true then  $P(k+1)$  is true



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Induction Hypothesis

If  $P(k)$  is true then  $P(k+1)$  is true

$$(P(1) \wedge \forall k(P(k) \rightarrow P(k+1))) \rightarrow \forall n P(n)$$

# Odd Powers Are Odd: Induction Idea

**Fact:** If  $m$  is odd and  $n$  is odd, then  $nm$  is odd.

**Proposition:** for an odd number  $m$ ,  $m^i$  is odd for all non-negative integer  $i$ .

$$\forall i \in \mathbb{Z} \quad \text{odd}(m^i)$$

Let  $P(i)$  be the proposition that  $m^i$  is odd.

$$\forall i \in \mathbb{Z} \quad P(i)$$

Idea of induction.

- $P(1)$  is true by definition.
- $P(2)$  is true by  $P(1)$  and the fact.
- $P(3)$  is true by  $P(2)$  and the fact.
- $P(i+1)$  is true by  $P(i)$  and the fact.
- So  $P(i)$  is true for all  $i$ .

# Divisibility by a Prime: Induction Idea

**Theorem.** Any integer  $n > 1$  is divisible by a prime number.

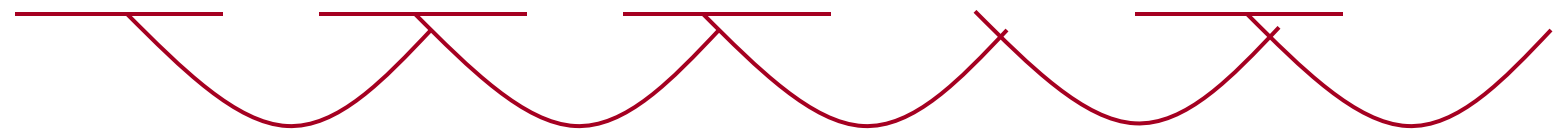
- Let  $n$  be an integer.
- If  $n$  is a prime number, then we are done.
- Otherwise,  $n = ab$ , both are smaller than  $n$ .
- If  $a$  or  $b$  is a prime number, then we are done.
- Otherwise,  $a = cd$ , both are smaller than  $a$ .
- If  $c$  or  $d$  is a prime number, then we are done.
- Otherwise, repeat this argument, since the numbers are getting smaller and smaller, this will eventually stop and we have found a prime factor of  $n$ .

Idea of induction.

# Idea of Induction

Objective: Prove  $\forall n \geq 0 \ P(n)$

This is to prove

$$\underbrace{P(0)} \wedge \underbrace{P(1)} \wedge \underbrace{P(2)} \wedge \dots \wedge \underbrace{P(n)} \dots$$


The idea of induction is to first prove  $P(0)$  unconditionally,  
then use  $P(0)$  to prove  $P(1)$   
then use  $P(1)$  to prove  $P(2)$   
and repeat this to infinity...

# The Induction Rule

0 and (from  $n$  to  $n+1$ ),

proves 0, 1, 2, 3, ....

Very easy  
to prove

Much easier to  
prove with  $P(n)$  as  
an assumption.

$$P(0), \forall n \in \mathbb{Z} P(n) \Rightarrow P(n+1)$$

valid rule

---

$$\forall m \in \mathbb{Z} P(m)$$

The point is to use the knowledge on smaller problems to solve bigger problems.



# Good and Bad of MI

## Good

- It can prove a conjecture once it has been made (and is true)
- Often simple

## Bad

- Cannot find/propose a new theorem
- Do not provide any insight as to why the theorem is true

# This Lecture

- The idea of mathematical induction
- Basic induction proofs (e.g. equality, inequality, property, etc)
- An interesting example
- A paradox

## Proving an Equality

$$r \neq 1 \quad \forall n \geq 1 \quad 1 + r + \dots + r^n = \frac{r^{n+1} - 1}{r - 1}$$

Let  $P(n)$  be the induction hypothesis that the statement is true for  $n$ .

Base case:  $P(1)$  is true because  $\text{LHS} = 1 + r = \text{RHS} = (r^2 - 1)/(r - 1) = r + 1$

Induction step: assume  $P(n)$  is true, prove  $P(n+1)$  is true.

That is, assuming:  $1 + r + \dots + r^n = \frac{r^{n+1} - 1}{r - 1}$

Want to prove:  $1 + r + \dots + r^n + r^{n+1} = \frac{r^{n+2} - 1}{r - 1}$

This is much easier to prove than proving it directly, because we already know the sum of the first  $n$  terms!



## Proving an Equality

$$r \neq 1 \quad \forall n \geq 1 \quad 1 + r + \dots + r^n = \frac{r^{n+1} - 1}{r - 1}$$

Let  $P(n)$  be the induction hypothesis that the statement is true for  $n$ .

Base case:  $P(1)$  is true

because  $\text{LHS} = 1 + r = \text{RHS} = (r^2 - 1)/(r - 1) = r + 1$

Induction step: assume  $P(n)$  is true, prove  $P(n+1)$  is true.

$$\begin{aligned} & 1 + r + \dots + r^n + r^{n+1} \\ &= \frac{r^{n+1} - 1}{r - 1} + r^{n+1} \quad \text{by induction} \\ &= \frac{r^{n+1} - 1 + r^{n+2} - r^{n+1}}{r - 1} \\ &= \frac{r^{n+2} - 1}{r - 1} \end{aligned}$$

## Proving an Equality

$$\forall n \geq 1 \quad 1^3 + 2^3 + \dots + n^3 = \left(\frac{n(n+1)}{2}\right)^2$$

Let  $P(n)$  be the induction hypothesis that the statement is true for  $n$ .

Base case:  $P(1)$  is true

Induction step: assume  $P(n)$  is true, prove  $P(n+1)$  is true.

$$\begin{aligned} & 1^3 + 2^3 + \dots + n^3 + (n+1)^3 \\ &= \left(\frac{n(n+1)}{2}\right)^2 + (n+1)^3 \quad \text{by induction} \\ &= (n+1)^2(n^2/4 + n + 1) \\ &= (n+1)^2\left(\frac{n^2 + 4n + 4}{4}\right) = \left(\frac{(n+1)(n+2)}{2}\right)^2 \end{aligned}$$

## Proving a Property

$$\forall n \geq 1, \quad 2^{2n} - 1 \text{ is divisible by } 3$$

Base Case ( $n = 1$ ):  $2^{2n} - 1 = 2^2 - 1 = 3$

Induction Step: Assume  $P(i)$  for some  $i \geq 1$  and prove  $P(i + 1)$ :

Assume  $2^{2i} - 1$  is divisible by 3, prove  $2^{2(i+1)} - 1$  is divisible by 3.

$$\begin{aligned} 2^{2(i+1)} - 1 &= 2^{2i+2} - 1 \\ &= 4 \cdot 2^{2i} - 1 \\ &= 3 \cdot 2^{2i} + \underbrace{2^{2i} - 1} \end{aligned}$$

Divisible by 3

Divisible by 3 by induction

# Proving a Property

$$\forall n \geq 2, \quad n^3 - n \text{ is divisible by } 6$$

Base Case ( $n = 2$ ):  $2^3 - 2 = 6$

Induction Step: Assume  $P(i)$  for some  $i \geq 2$  and prove  $P(i + 1)$ :

Assume  $n^3 - n$  is divisible by 6

Prove  $(n + 1)^3 - (n + 1)$  is divisible by 6.

$$\begin{aligned}(n + 1)^3 - (n + 1) &= (n^3 + 3n^2 + 3n + 1) - (n + 1) \\ &= (n^3 - n) + 3(n^2 + n)\end{aligned}$$

Divisible by 6  
by induction

Divisible by 2  
by case analysis

HOW  
?

# $n^2 + n$ is divisible by 2 (Case Analysis)

- Say  $n$  is odd:
  - Recall: for an odd number  $m$ ,  $m^i$  is odd for all non-negative integer  $i$ .
  - So,  $n^2$  is odd;  $n$  is also odd
  - $n^2 + n$  is sum of two odd numbers  $\Rightarrow$  even
- Say  $n$  is even
  - Easy...

## Proving an Inequality

$$\forall n \geq 3, \quad 2n + 1 < 2^n$$

Base Case ( $n = 3$ ):  $2n + 1 = 7 < 2^n = 2^3 = 8$

Induction Step: Assume  $P(i)$  for some  $i \geq 3$  and prove  $P(i + 1)$ :

Assume  $2i + 1 < 2^i$ , prove  $2(i + 1) + 1 < 2^{(i+1)}$

$$2(i + 1) + 1 = \underline{2i + 1} + 2$$

$$< 2^i + 2 \quad \text{by induction}$$

$$< 2^i + 2^i \quad \text{since } i \geq 3, \text{ we must have } 2 < 2^i$$

$$= 2^{(i+1)}$$

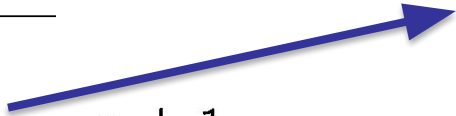
## Proving an Inequality

$$\forall n \geq 2, \quad \frac{1}{\sqrt{1}} + \frac{1}{\sqrt{2}} + \dots + \frac{1}{\sqrt{n}} > \sqrt{n}$$

Base Case ( $n = 2$ ) is true [ $1/\sqrt{1} + 1/\sqrt{2} = 1.71 > 1.41 = \sqrt{2}$ ]

Induction Step: Assume  $P(i)$  for some  $i \geq 2$  and prove  $P(i + 1)$ :

$$\begin{aligned} & \frac{1}{\sqrt{1}} + \frac{1}{\sqrt{2}} + \dots + \frac{1}{\sqrt{n}} + \frac{1}{\sqrt{n+1}} \\ & > \sqrt{n} + \frac{1}{\sqrt{n+1}} \quad \text{by induction} \\ & = \frac{\sqrt{n}\sqrt{n+1} + 1}{\sqrt{n+1}} \\ & > \frac{\sqrt{n}\sqrt{n} + 1}{\sqrt{n+1}} = \frac{n+1}{\sqrt{n+1}} \\ & = \sqrt{n+1} \end{aligned}$$

  $\sqrt{n+1} > \sqrt{n}$

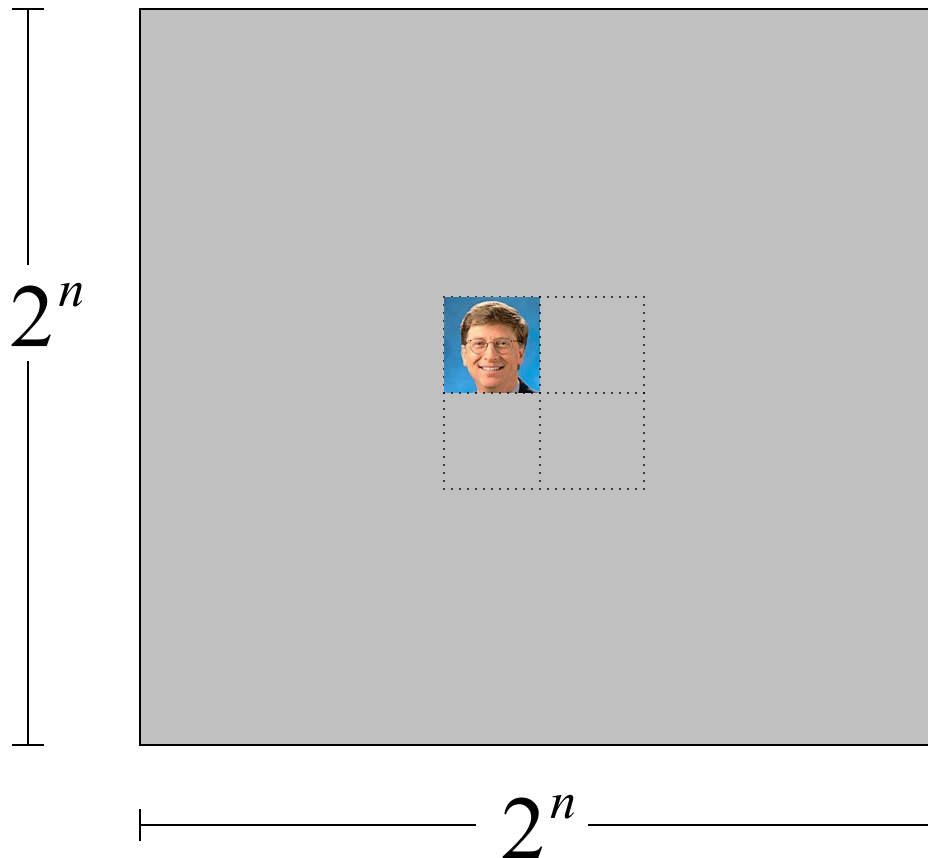
# This Lecture

- The idea of mathematical induction
- Basic induction proofs (e.g. equality, inequality, property, etc)
- An interesting example
- A paradox



# Puzzle

**Goal:** tile the squares, except one in the middle for Bill.

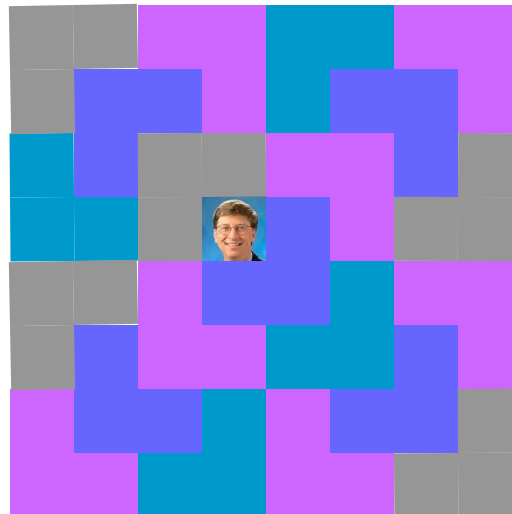


# Puzzle

There are only trominos (L-shaped tiles) covering three squares:



For example, for 8 x 8 puzzle might tile for Bill this way:



# Puzzle

**Theorem:** For any  $2^n \times 2^n$  puzzle, there is a tiling with Bill in the middle.

Did you remember that we proved  $2^{2n} - 1$  is divisible by 3?

Proof: (by induction on  $n$ )

$P(n) ::=$  can tile  $2^n \times 2^n$  with Bill in middle.

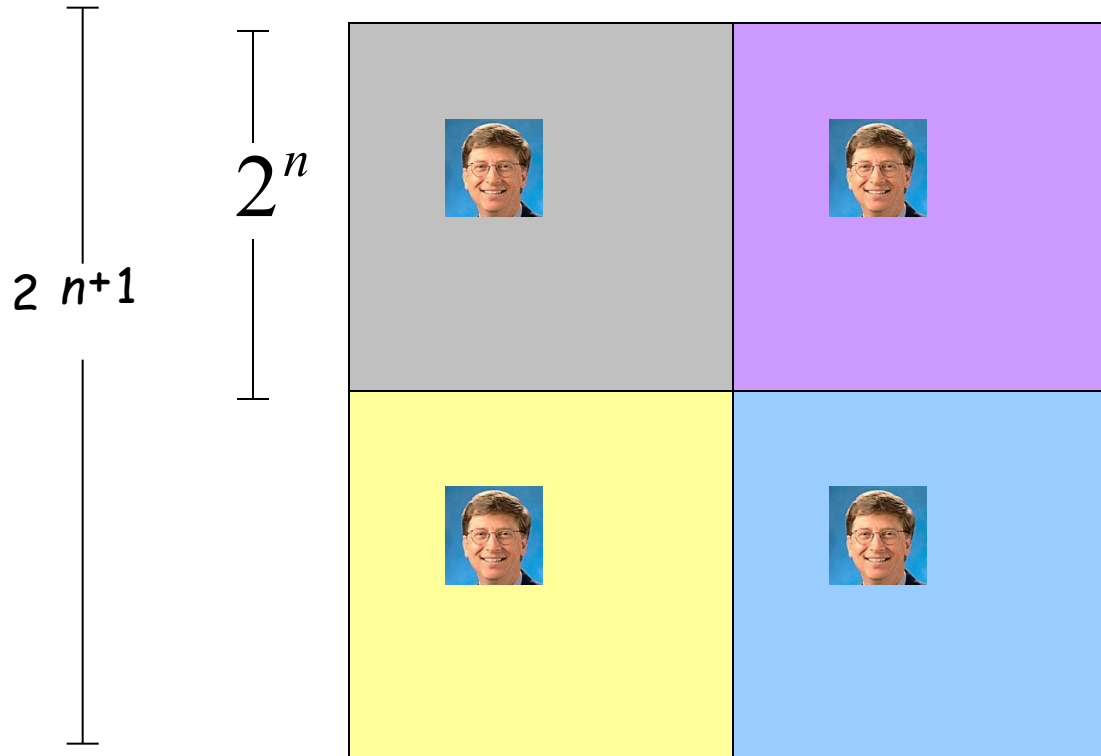
Base case: ( $n=0$ )



(no tiles needed)

# Puzzle

Induction step: assume can tile  $2^n \times 2^n$ ,  
prove can handle  $2^{n+1} \times 2^{n+1}$ .



# Puzzle

The new idea:

A stronger property

Prove that we can always find a tiling with Bill *anywhere*.

**Theorem B:** For any  $2^n \times 2^n$  plaza, there is a tiling with Bill *anywhere*.

Clearly Theorem B implies Theorem.

**Theorem:** For any  $2^n \times 2^n$  plaza, there is a tiling with Bill in the middle.

# Puzzle

**Theorem B:** For any  $2^n \times 2^n$  plaza, there is a tiling with Bill anywhere.

Proof: (by induction on  $n$ )

$P(n) ::=$  can tile  $2^n \times 2^n$  with Bill anywhere.

Base case: ( $n=0$ )



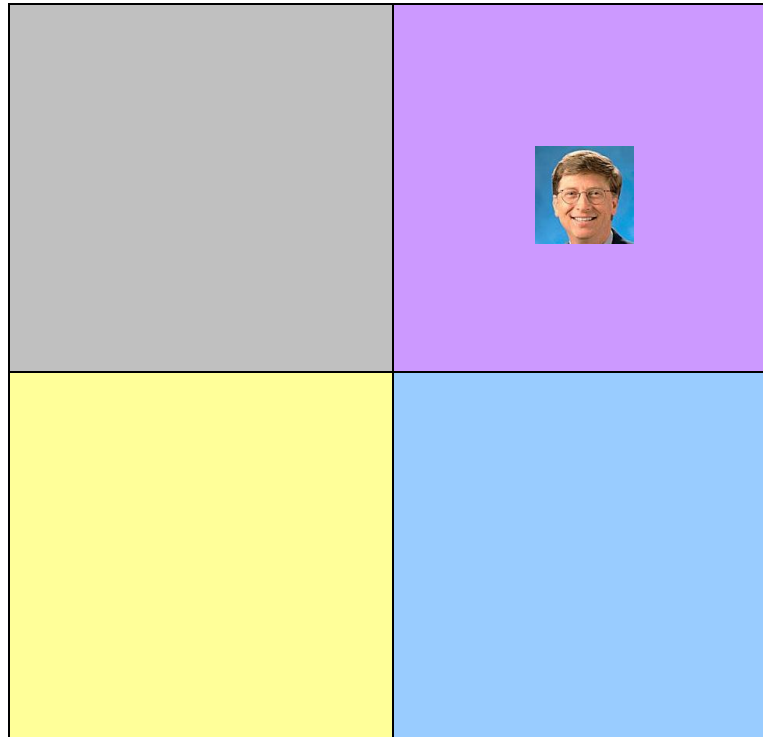
(no tiles needed)

# Puzzle

Induction step:

Assume we can get Bill *anywhere in*  $2^n \times 2^n$ .

Prove we can get Bill anywhere in  $2^{n+1} \times 2^{n+1}$ .

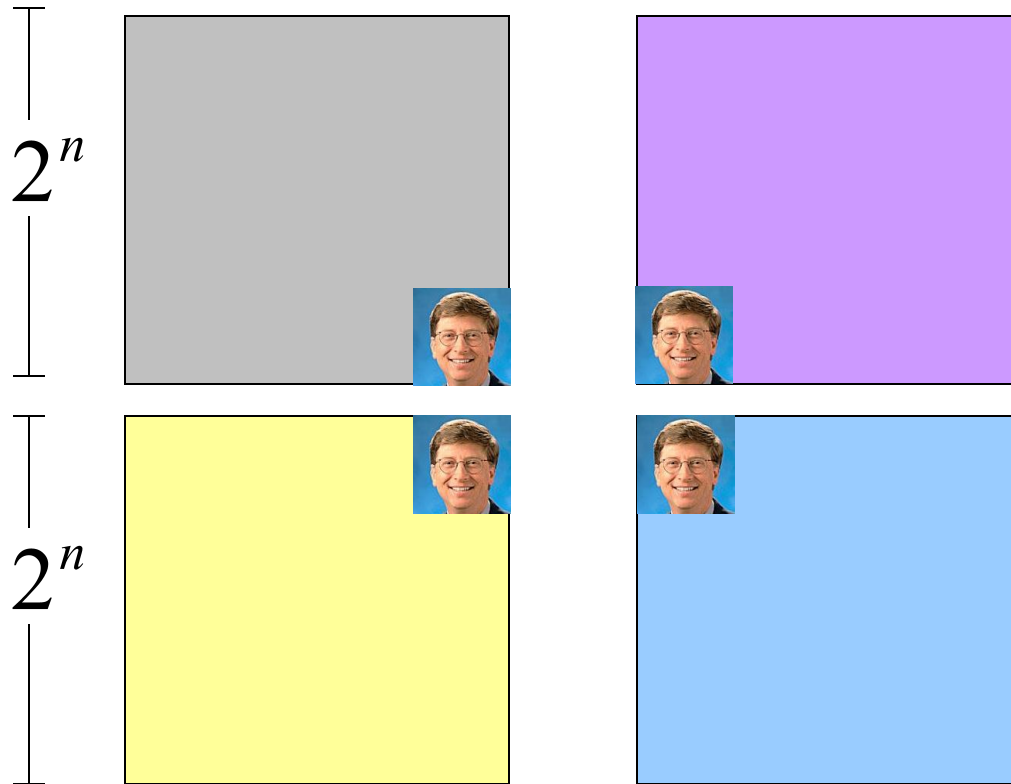


# Puzzle

Induction step:

Assume we can get Bill *anywhere* in  $2^n \times 2^n$ .

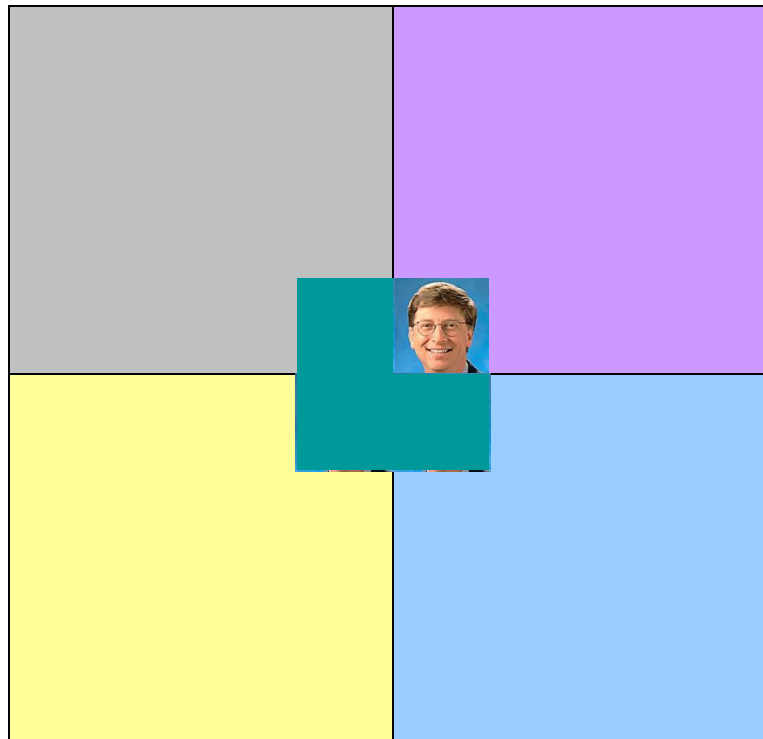
Prove we can get Bill anywhere in  $2^{n+1} \times 2^{n+1}$ .





# Puzzle

**Method:** Now group the squares together,  
and fill the center with a tile.



**Done!**

# This Lecture

- The idea of mathematical induction
- Basic induction proofs (e.g. equality, inequality, property, etc)
- An interesting example
- A paradox

# Paradox

*Theorem:* All horses have the same color.

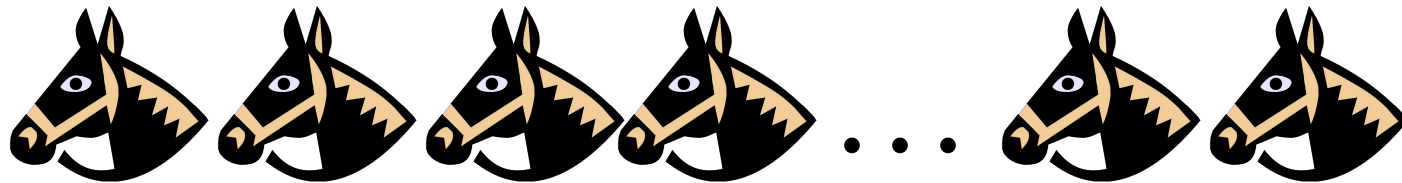
*Proof:* (by induction on  $n$ )

Induction hypothesis:

$P(n) ::=$  any set of  $n$  horses have the same color

Base case ( $n=0$ ):

No horses so *obviously* true!

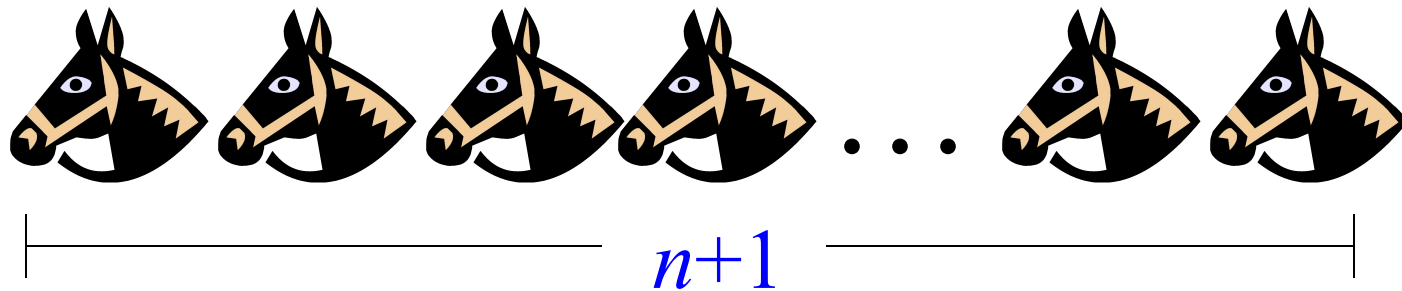


# Paradox

(Inductive case)

Assume any  $n$  horses have the same color.

Prove that any  $n+1$  horses have the same color.

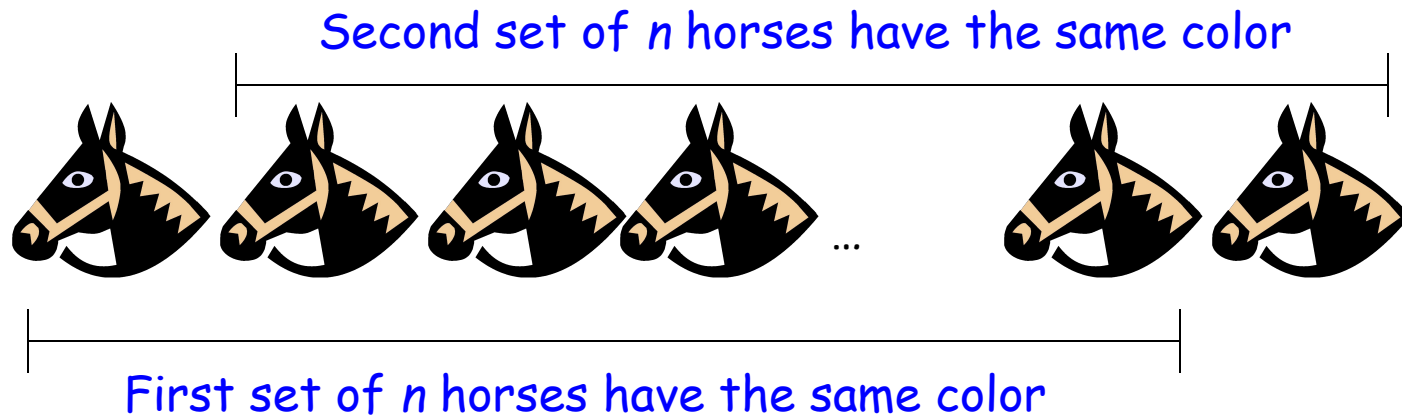


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(Inductive case)

Assume any  $n$  horses have the same color.

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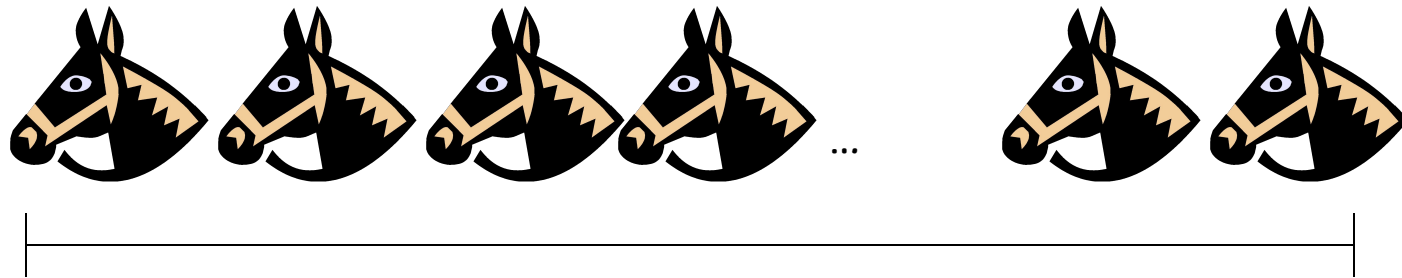


# Paradox

(Inductive case)

Assume any  $n$  horses have the same color.

Prove that any  $n+1$  horses have the same color.

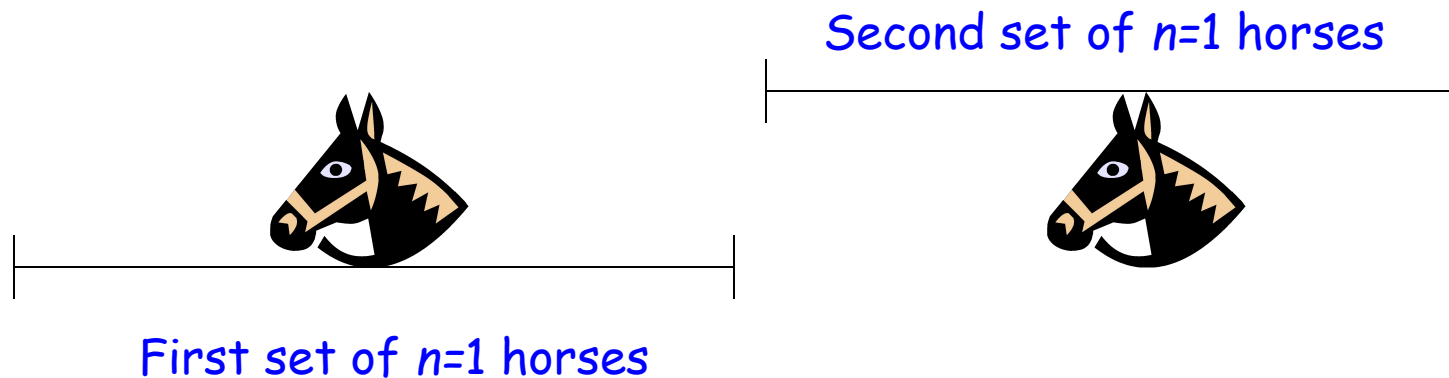


Therefore the set of  $n+1$  have the same color!

# Paradox

What is wrong?  $n = 1$

Proof that  $P(n) \rightarrow P(n+1)$   
is false if  $n = 1$ , because the two  
horse groups do not overlap.



(But proof works for all  $n \neq 1$ )

*The main problem is that you have assumed an overlap between any two sets;  
This is NOT true*

# Quick Summary

You should understand the principle of mathematical induction well,  
and do basic induction proofs like

- proving equality
- proving inequality
- proving property

Mathematical induction has a wide range of applications in computer science.

In the next lecture we will see more applications and more techniques.